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# A Novel Delay-Line Time-to-Digital Converter with Logic-Driven Time Encoding

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***Abstract* - A Time-to-Digital Converter (TDC) employing 45 nm CMOS technology is proposed in this work. It uses a singular delay-line architecture for accurate measurement of time intervals in digital circuits. The architecture employs 5 current-starved delay cells, each exhibiting a propagation delay of 245 ps, coupled with six edge-triggered D flip-flops to capture signal transitions and generate a thermometer code. A custom digital encoder converts this thermometer coded sequence into a 3-bit output in binary. The design requires low power, which makes it suitable for on chip performance monitoring. The integrated TDC architecture supports applications such as electronic control unit synchronization and time-of-flight measurements in sensors for safety critical systems.**

***Keywords - TDC, delay line, current controlled inverter, encoder, thermometer code.***

## I. INTRODUCTION

TDCs calculate the difference in time between two events and convert it into digitized data. By using reference clocks or delay elements, they divide time into discrete steps and represent the temporal difference in the start and stop signals in digitized form. Their scalability, low power consumption, and CMOS integration make them indispensable in LiDAR, time-of-flight imaging, oscilloscopes, and on chip monitoring. Immense research has been conducted on TDC architectures. In [1] TDCs and DTCs are studied in DLLs and counters, [2] explored flash, vernier line, vernier delay ring, and replica delay line architectures, while [3] shows comparison of sampling based, noise shaping, stochastic, and multi phase clock-based architectures. Delay-line TDCs are widely adopted for their simplicity.

In this work, we present a delay line based TDC implemented in 45 nm CMOS technology with five delay cells and six D flip-flops. While conventional TDCs focus on timing measurement, our design also functions as a test structure for assessing transistor aging, providing valuable insights for reliability studies in advanced digital systems.

## II. LITERATURE SURVEY

The design and optimization of TDCs has advanced greatly in the past two decades, due to the demand for high resolution, low power, and compact timing circuits in applications such as data converters, LiDAR, high speed communication, and aging analysis.

Chen Yao [4] presented high resolution TDCs in 65 nm CMOS tailored for phase detection in an All Digital PLL operating with a 5 GHz DCO and 20 MHz reference. In the same year, Moazedi, Abrishamifar, and Sodagar [5] described a current-starved delay element using a four-stage delay line, pseudo differential in nature, used for analog or mixed DLLs, which used body feed technique to widen the control voltage range.

Tancock, Arabul, and Dahnoun [6] gave a thorough review of TDC architectures used on ASICs and FPGAs, emphasizing on successive approximation, algorithmic TDCs, wave union launchers, and FPGA based series TDCs. Ryan Scott, Wei Jiang, and Jamal Deen [7] explored TDC architectures for biomedical imaging purpose, focusing on integration with SPAD imagers and dSiPMs and their design trade offs unique to medical imaging.

Hassan et al. [8] focused on low power TDCs for pixel level applications, implementing a 65 nm design with 180 ps resolution, 119 µW average power, and 644 µm² area using a gated ring oscillator.

Early works emphasized robust time mode processing and high resolution phase detection, while recent trends as demonstrated by [9] focuses on aggressive power reduction through architectural innovation such as calibrated single delay lines for physics detectors, and signal continuity aware counting for imaging systems. The proposed implementation occupies ultra low power (3.79 µW), structural simplicity, and transistor aging test capability at moderate resolution in a 45 nm process is not addressed by either of the above works.

## III. METHODOLOGY

This work is done on the basis of a tapped delay line circuit architecture. It is used to measure the difference in timings of two input signals, a delayed one and a reference one. The reference signal given as input, propagates through a chain of delay units, while the delayed signal works like a clock triggering the D flip-flops. State of signal for each flip-flop is captured at different points along the delay line, creating a thermometer coded output that shows how much the reference signal traveled when the delayed signal arrived.

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### A. Design of Delay Element

The TDC delay unit uses a current-starved inverter delay cell followed by a non-inverting buffer for fine timing control and creating a digital logic level output. The delay for an individual delay cell is calculated using equations (1) and (2).

a) $$I_D = \frac{1}{2} * \mu * C_{ox} * \frac{W}{L} * (V_{GS} - V_T)^2 \quad (1)$$

b) $$t_{delay} = \frac{V_{DD} \cdot C_L}{I_D} \quad (2)$$

Varying the transistor conditions delay changes. These current limiting transistors have their gates connected to external DC biasing voltages. These voltages set the drive current for the inverter, effectively controlling the traversing delay through the cell. In addition, a load capacitor of 4.7 fF is placed at the output of the delay cell to model the fan out loading, ensuring that the simulated delay matches real world scenarios.

Using a technology node of 45 nm scale, the design parameters considered for development of internal circuitry state that the total width of 45 nm and length of 120 nm for both PMOS and NMOS.

The external bias voltages mentioned in Table I. are selected to ensure controlled and symmetric current starvation by limiting both the PMOS and NMOS current paths. With Vbp set lower than Vbn, the PMOS remains weakly ON and the NMOS moderately ON, keeping both transistors in saturation but with restricted drive strength thus slowing both rising and falling edges.

### B. Design of D Flip-Flop

This CMOS logic based D flip-flop, ensures rising edge-triggered operation for high speed sampling required in TDCs.

### C. Integration of Delay line and Flip-flops

In the equation (3), N denotes the count of delay cells, Tlsb gives the measured delay of each unit, T is the time period.

$$T = N \cdot T_{LSB} \quad (3)$$

The integration of delay line and flip-flops connects the five delay stages in series, with each stage feeding into a corresponding D flip-flop. The reference input signal is applied to the first delay stage, while the other delayed signal is used as a synchronous clock across all flip-flops. This configuration ensures precise measurement of delay between the two signals by triggering the flip-flops at successive edges along the delay chain. The inclusion of a sixth flip-flop confirms sampling of the final delay tap enabling accurate edge detection.

### D. Design of 6:3 Binary Encoder

The 6:3 encoder converts the thermometer code obtained by multiple D flip-flop outputs from the delay line into a binary 3-bit code representing the time interval. A combinational circuit is used to depict the delay based on position of time difference occurrence.

The encoder is designed to identify the position of the first occurrence of digital logic LOW in the delay line output sequence and map it to a binary code using minimal logic. This logic is explicitly expressed in terms of boolean algebra to ensure accurate edge detection. The encoder logic is derived manually to minimize delay, and is capable of handling metastability by defaulting to a fixed code when all bits are digital logic HIGH. The logic equations for the output bits with MSB Y2, Y1, Y0 of the encoder:

$$Y_2 = D_1 D_2 \underline{D_3}\ \underline{D_4}\ \underline{D_5}\ \underline{D_6} + \underline{D_1}\ \underline{D_2}\ \underline{D_3}\ \underline{D_4}\ \underline{D_5}\ \underline{D_6} + D_1 D_2 D_3 \underline{D_5}\ \underline{D_6} \quad (4)$$

$$Y_1 = D_1 D_2 \underline{D_3}\ \underline{D_4}\ \underline{D_5}\ \underline{D_6} + D_1 D_2 D_3 D_4 D_5 \underline{D_6} + D_1 \underline{D_2}\ \underline{D_3}\ \underline{D_4}\ \underline{D_5}\ \underline{D_6} \quad (5)$$

$$Y_0 = D_1 D_2 D_3 \underline{D_5}\ \underline{D_6} + D_1 D_2 D_3 D_4 D_5 \underline{D_6} \quad (6)$$

Each equation corresponds to a particular transition pattern in the thermometer sequence. The use of both asserted and negated D flip-flop outputs allows the encoder to precisely detect where the first falling edge occurs, which represents the arrival of the input signal edge at a particular stage in the delay line. Table I. enlists the entire thermometer code encoding for the TDC circuit output.

TABLE I. ENCODER OUTPUT

| Input to encoder [ Q1, Q2, Q3, Q4, Q5, Q6 ] | Encoder Output (Y2, Y1, Y0) |
|---|---|
| 111111 | 000 |
| 111110 | 110 |
| 111100 | 101 |
| 111000 | 100 |
| 110000 | 011 |
| 100000 | 010 |
| 000000 | 001 |

### E. TDC design

In Fig. 1, the complete TDC designing process is shown, which includes integration of the delay chain, flip-flops, and the encoder.

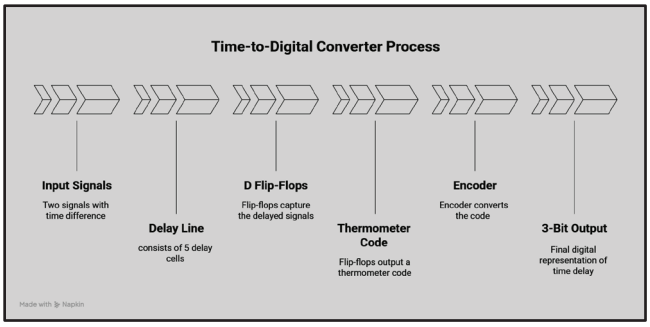



Fig. 1. Block diagram of TDC

Each signal output from the flip-flop is buffered before reaching the encoder to maintain signal integrity, preventing loading effects and ensuring smooth transitions. The 3-bit output of the encoder is also buffered to stabilize the final digital code proper output.

## IV. IMPLEMENTATION

The presented TDC circuit is implemented on the Cadence Virtuoso. This section provides the implementation of different blocks of the circuit along with their integration.

### A. Delay Cell

In Fig. 2, a delay element designed using the Cadence Design Tool is illustrated.



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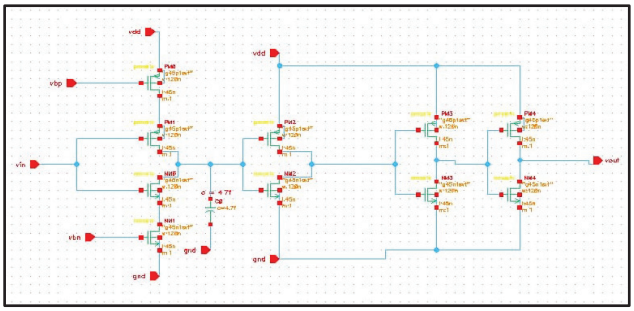


Fig. 2. Current Starved Delay element schematic

We have four different input terminals, constituting input and bias signals. The internal inverter is given the input signal to both of its transistor gates. To limit current flow, an additional PMOS transistor is inserted between the supply voltage and the inverter's PMOS pull up device, while an NMOS transistor is placed between the inverter's NMOS pull down and ground.

### *B. D Flip-Flop*

Fig. 3 demonstrates a one bit D flip-flop constructed using interconnected NAND gates.

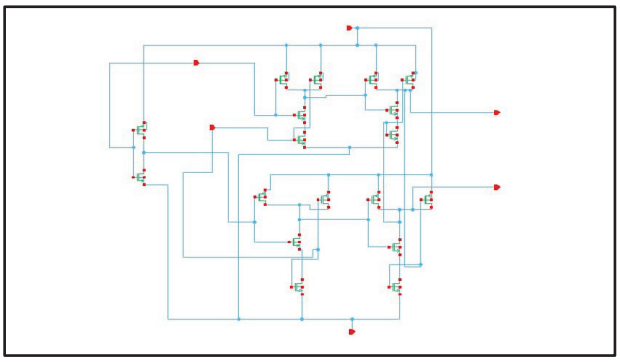


Fig. 3. Rising Edge-Triggered D Flip FLop

### *C. Integrated Delay line and Flip-Flops*

Each delay stage output is routed to data input of a flip-flop, with the initial delay cell output connected to the second flip-flop, establishing a sequential tap arrangement with a common clock input, which is derived from the delayed input signal. In Fig. 4, the integrated architecture of the flip-flop and delay line is shown as a schematic representation. The first delay stage gets the input signal, while the other delayed signal is used as a synchronous clock across all flip-flops.

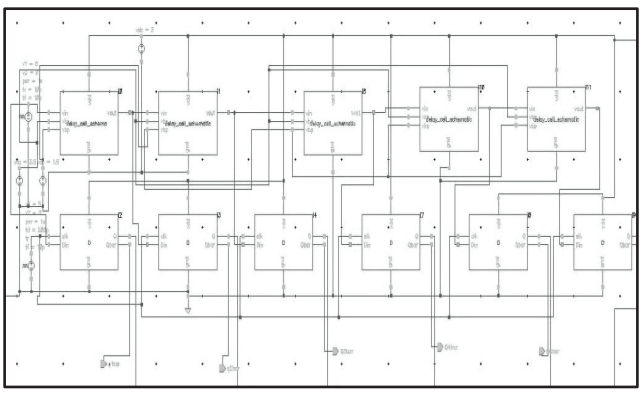


Fig. 4. Schematic of concatenated Delay cells and flip-flops

Parameters used during the simulation of this block are summarized in Table II.

TABLE II. DESIGN AND SIMULATION PARAMETERS OF INTEGRATED TDC CIRCUIT

| Cell View Name | Use case | Properties (45 nm) |
|---|---|---|
| Vpulse | Reference Input Signal | voltage 1 = 1 V<br>fall/rise Time = 5 ps<br>period = 20 ns |
| Vpulse | Delayed Clocked Signal | voltage 1 = 1V<br>period = 20 ns<br>delay time = 150 ps<br>rise/fall Time = 5 ps |
| Vdc | Supply Voltage | DC voltage = 1 V |
| Vdc | Vbp, Vbn (External Biasing voltage) | (Vbp)DC voltage = 0.15 V<br>(Vbn)DC voltage = 0.85 V |

### *D. Binary Encoder*

Fig. 5. consists of the schematic of 6:3 Binary Encoder which gives a 3-bit output from the 6-bit thermometer code generated. The basis for the encoder design is the combinational logic developed. The Input is Q0 to Q5 for the output of Y0, Y1 and Y2.

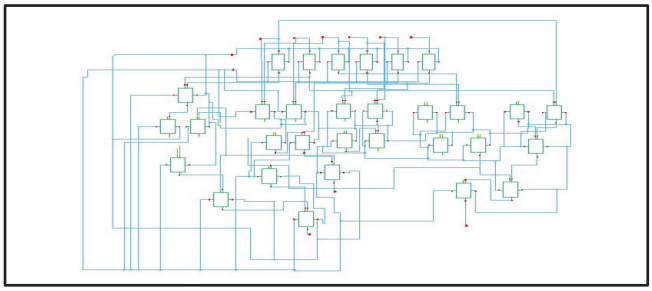


Fig. 5. Schematic of 6:3 Binary Encoder

### *E. TDC Implementation*

In Fig. 6, the entire TDC architecture schematic is shown. The delay line, D flip-flops, and encoder are integrated in hierarchical layout. This concludes the circuit level methodology, forming the foundation for further analysis.

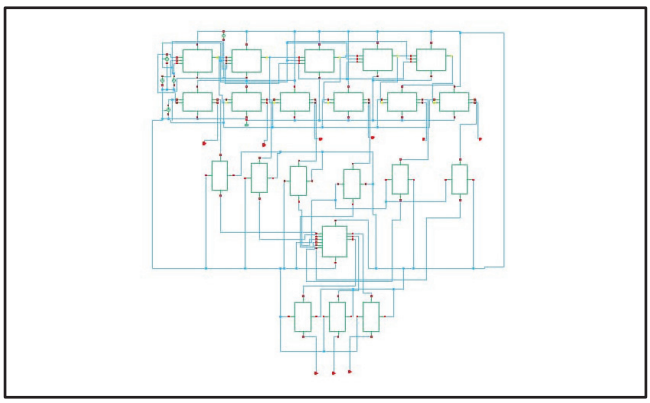


Fig. 6. Schematic of TDC



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## V. Results and Discussion

The simulation of the TDC in this work is done in GPDK 45 nm technology using Cadence Virtuoso. The section presents the key simulation results, including the performance of all the present components along with the TDC's encoded outputs, and the evaluation of power and metrics. All simulations are performed under nominal conditions with a 1V supply voltage.

### *A. Delay Cell Characterization*

To validate the performance of the current-starved inverter based delay element, transient simulations are conducted with various biasing voltages and load conditions. For the delay line, input and output behaviour is demonstrated in Fig. 7, under a capacitive load of 4.7 fF. The applied DC bias voltages are Vbp (0.85 V) and Vbn (0.15 V) according to the designed methodology.

The obtained propagation delay for the above conditions is 245 ps. This delay was calculated as the critical temporal difference between input rising edge and the delayed output.

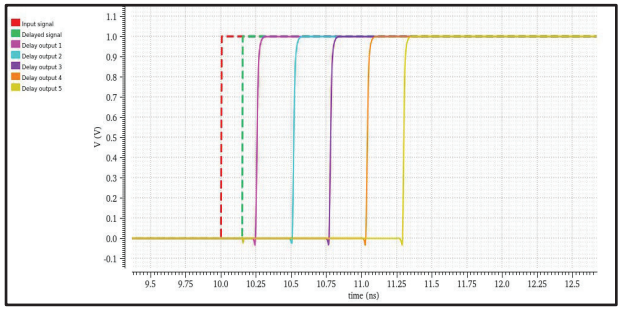



Fig. 7. Output of Concatenated Delay Stages

### *B. Delay Line Output*

An overall delay range of 1290 ps, gives a time resolution of 245 ps per tap. As shown in Fig. 7, at 10 ns the rising edge occurs for the input. Hence, the delay stages listed in Table III. are activated sequentially, synchronizing with this input transition.

TABLE III. Comparison of Expected and Observed Delays cross Delay Stages

| Delay stages | Expected delay | Observed delay |
|---|---|---|
| Stage 1 | 245 ps | 245.68 ps |
| Stage 2 | 490 ps | 505.57 ps |
| Stage 3 | 735 ps | 768.92 ps |
| Stage 4 | 980 ps | 1030.5 ps |
| Stage 5 | 1225 ps | 1289.2 ps |

As outlined before in Equation (3), the delay increases additively at each successive stage, forming a cumulative sequence of expected delays, as shown in Table III. comparing these theoretical values with the observed delays.

As displayed in Fig. 8, the input signal's rising edge at 10 ns triggers propagation through the delay chain. The delay line's output is captured at each clock edge of the flip-flop .In this case, a time difference of approximately 150 ps between the reference and delayed signals falls within the range of the first delay cell.

As a result, at the second flip-flop's output, the first HIGH to LOW transition is observed, indicating that the delay lies within the window of 0 ps to 245 ps, which corresponds to the resolution of the first stage. This transition effectively marks the point in the delay chain where the signals diverge. Fig. 8. successfully demonstrated a 6-bit code of 100000 on the time instance of delayed signal clock edge.

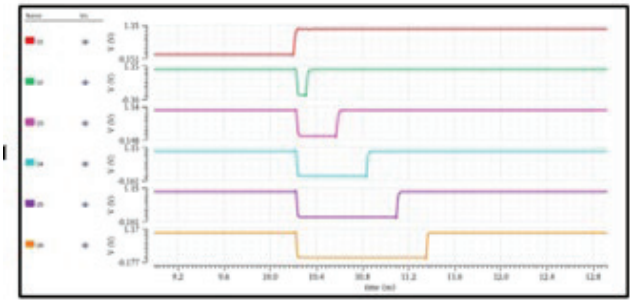


Fig. 8. Output of Delay Line Flip Flops with 150 ps Time Difference Measurement

To further support the observed behavior, an additional case with a time difference of 370 ps is presented in Fig 9. resulting in the output code 110000. This consistent pattern denotes the location of the first HIGH to LOW transition corresponding to the specific delay stage. This makes the delay chain approach more clear and reliable.

### *C. Thermometer Code and Encoding*

The delay line outputs are sampled by six D flip-flops triggered by a delayed signal. The resulting thermometer code is shown in Fig. 9, where each flip-flop output captures the logic level of the delay node at the arrival of the delayed clock. For better understanding, consider a sample code matrix (7).

$$[Q_1\ Q_2\ Q_3\ Q_4\ Q_5\ Q_6] = [1\quad 1\quad 1\quad 1\quad 0\quad 0\ ] \quad (7)$$

The 6-bit code is encoded to binary 101 by the custom thermometer to binary encoder, as verified in Fig. 9. The encoder output changes precisely with the transition point in the thermometer code, confirming correct operation of edge detection logic.

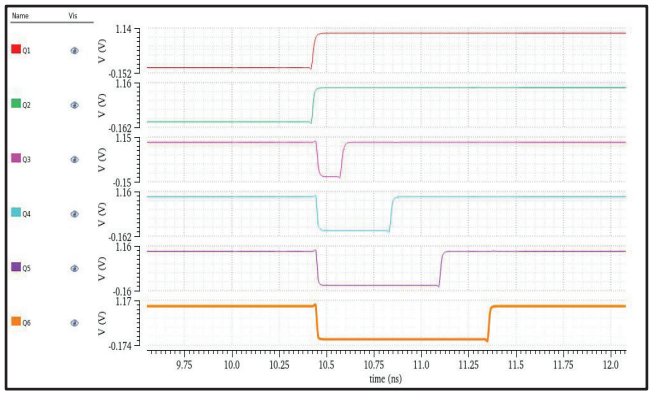



Fig. 9. Output of Delay Line Flip Flops with 370 ps Time Difference Measurement

As illustrated in Fig. 10, these waveforms are transformed into a 3-bit binary representation by identifying the position at


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which the first HIGH-to-LOW transition occurs in the corresponding thermometer code.

For instance at a time difference of 150 ps referencing back to Fig 8. The transition at Q2 indicates the delay falls within the first delay cell and is represented by the code 010.

As the delay increases and transition occurs at later flip-flops, the encoder generates corresponding codes such as 011, 100, 101, and so on for each delay range. Time difference is effectively quantized into discrete digital levels based on which flip-flop first detects the edge.

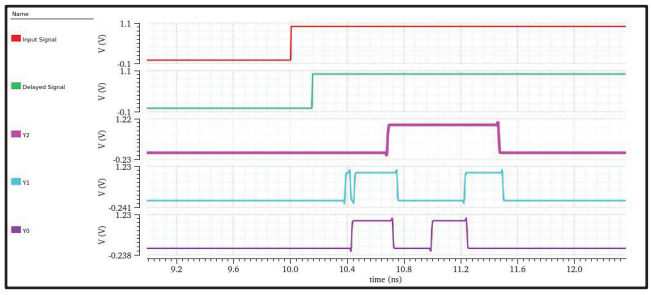



Fig. 10. Output Graph of 3-bit Binary Encoder for 150 ps Time Delay

For a time difference of approximately 370 ps, the first HIGH to LOW transition shifts to later flip-flops, producing output codes such as 011, 100, 101, and 110, as shown in Fig. 11.

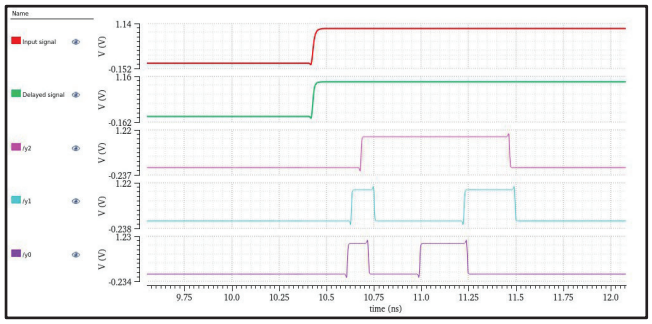



Fig. 11. Output Graph of 3-bit Binary Encoder for 370 ps Time Delay

The corresponding delay range is illustrated in the timing diagram of Fig. 9.

This progression confirms that increasing delay results in encoder outputs that quantize the time difference into discrete intervals defined by the delay.

### *D. Performance analysis*

1) *Power optimization*- Average power consumption is estimated using Cadence ADE-L by measuring the integrated current drawn over one period of input signal, found to be 3.79 µW. The TDC exhibits low-power operation, benefiting from current-starved inverters and minimal logic depth in the encoder.

2) *Linearity and resolution*- Linearity is evaluated by measuring delay values per stage across multiple test cases. A nearly uniform 245 ps delay per cell is maintained with minor deviation ± 35 ps due to transistor mismatch and load variations.

## VI. CONCLUSION

The work elaborated in this paper introduces a novel TDC design, with delay-line circuit as its basis, that is implemented in 45 nm CMOS technology providing better accuracy and performance. The proposed implementation distinguishes itself from existing designs by incorporating a custom developed encoder architecture that enables faster and more accurate conversion of thermometer code into binary form. Table IV provides a comparison among the previously designed TDC architectures which makes this work reliable in time measurement with reduced power consumption. Enhanced resolution is achieved through the optimized delay elements in this work.

Furthermore, the design exhibits scalability for applications involving fine grained timing while remaining resilient to aging effects and process variations. The results demonstrate that the proposed architecture is well suited for integration into high performance digital systems, including communication platforms, high speed data converters, and time critical SoC applications, owing to its low power consumption and high timing accuracy. Additionally, its fast response and low power consumption make it a viable solution for clock skew detection and synchronization in automotive ECU networks.

## ACKNOWLEDGEMENT

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TABLE IV. COMPARISON OF TDC ACROSS DIFFERENT RESEARCH PAPER

| Reference | Technology | Architecture | Resolution | Power | Output Bits | Primary Application |
|---|---|---|---|---|---|---|
| Moazedi et al. [5], ICEE 2011 | Standard CMOS | Pseudo-diff. current-starved DLL | ~100 ps range | Low | Analog | Mixed-signal DLLs |
| Hassan et al. [8], IICM 2022 | 65 nm CMOS | Gated Ring Oscillator | 180 ps | 119 µW avg. | N/A | Pixel-level imaging |
| Zhang et al. [10], IEEE TNS 2021 | 65 nm CMOS | Single delay line, dual-strobe self-calibration | 17.8 ps/bin | <200 µW/pixel | N/A | HL-LHC TOA+TOT particle detection |
| Li & Gao [11], Micromachines 2024 | 110 nm CMOS | Dual-row delay chain, SS ADC optimised | 5-bit quantisation | 60–93 µW | 5-bit | High-speed CMOS image sensors |
| This Work | 45 nm GPDK CMOS | Current-starved delay line + custom 6:3 encoder | 245 ps/tap | 3.79 µW | 3-bit | On-chip monitoring, ToF, ECU sync |



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